

Lecture 1

Why and what is probability?

Probability is used to model uncertainty we see in nature.

Any random experiment has uncertain outcomes and we want to find how likely is it to observe a certain outcome.

Probability theory is a mathematical framework to analyze and understand the random experiment.

Applications of probability.

- ① Statistics
- ② Machine Learning / Reinforcement Learning
- ③ Communications
- ④ Theoretical Computer Science

Probabilistic algorithms

Probabilistic method (Combinatorics,
coding and information theory)

Different Approaches to probability

① Classical approach: To find probability of observing an event A

$$P(A) = \frac{\# \text{ of outcomes that favour } A}{\text{total } \# \text{ of outcomes that are possible.}}$$

Example: Roll two dice, what is the probability that sum is 9.
 $A :=$

① outcome is defined as the sum that is observed.

2, 3, 4, ..., 12 are the possible outcomes.

$$P(A) = \frac{1}{11}$$

⑥ Suppose outcomes are the values seen on both the dice then

(1, 1)	(1, 2)	-	-	(1, 6)
(2, 1)		-	-	(2, 6)
.				
.				
(6, 1)		-	-	(6, 6)

total # of outcomes = 36

favourable outcomes : = (3, 6) (4, 5)
(5, 4) (6, 3)

$$P(A) = \frac{4}{36} = \frac{1}{9}.$$

this definition will not help us if the die is not fair.

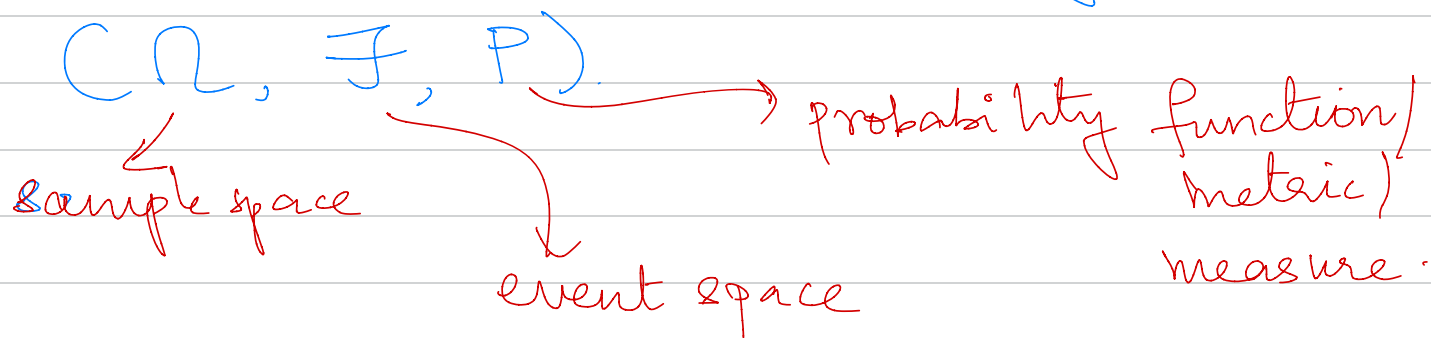
② Frequentist approach: Run an experiment ^{infinite} number of times.

$$P(A) = \lim_{n \rightarrow \infty} \frac{a_n}{n}.$$

a_n = # of times you observed an outcome favourable to event A .

③ Axiomatic approach to probability (Kolmogorov).

Probabilistic model is described through



Ω : Sample space : collection of all the experiment outcomes.

① In the earlier example of two dice roll it is

$$\Omega = \{(1,1) \quad \dots \quad (6,6)\} \quad |\Omega| = 36.$$

② If single die roll

$$\Omega = \{1, 2, \dots, 6\} \quad |\Omega| = 6.$$

③ Toss a coin till you see a head.

$$\Omega = \{H, TH, TTH, TTTH, \dots\}$$

$\mathcal{F}$: event space.

set containing events.

events are subsets of the sample space Ω .

If $A \in \mathcal{F}$ then $A \subseteq \Omega$.

Examples

(a) Suppose $\Omega = \{1, 2, 3, \dots, 6\}$

Single die roll.

$A :=$ even outcome.
 $= \{2, 4, 6\}$

$A^c = \{1, 3, 5\}$.

(a.1) $\mathcal{F} = \{\emptyset, A, A^c, \Omega\}$

(a.2) $\mathcal{F} = \{\emptyset, \Omega, \{1\}, \{2\}, \{3\}, \dots, \{1, 2\}, \dots\}$
all possible subsets of Ω . $\leftarrow \mathcal{F} = 2^\Omega$

$$P : \mathbb{I} \longrightarrow [0, 1].$$

Basics on sets and operations over them -

Set is collection of objects referred to as elements.

$x \in S \Rightarrow x$ is an element in set S .

$$\textcircled{1} A^c = \{x \in \Omega \mid x \notin A\}.$$

$$\textcircled{2} A \cup B = \{x \in \Omega \mid x \in A \text{ or } x \in B\}$$

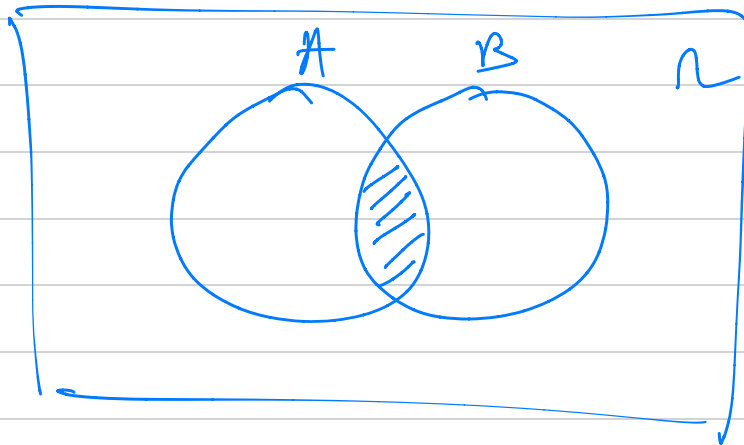
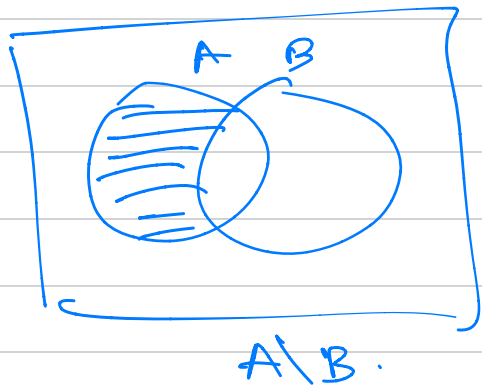
$$\textcircled{3} A \cap B = \{x \in \Omega \mid x \in A \text{ and } x \in B\}$$

$$\textcircled{4} A \setminus B = \{x \in \Omega \mid x \in A, x \notin B\}$$

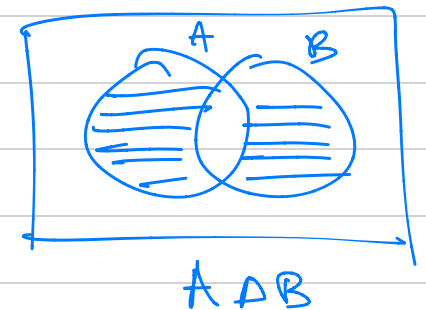
$$\textcircled{5} A \Delta B = (A \setminus B) \cup (B \setminus A)$$

$$A_1 \cup \dots \cup A_n = \{x \in \Omega \mid x \in A_i \text{ for some } i \in \{1, 2, \dots, n\}\}$$

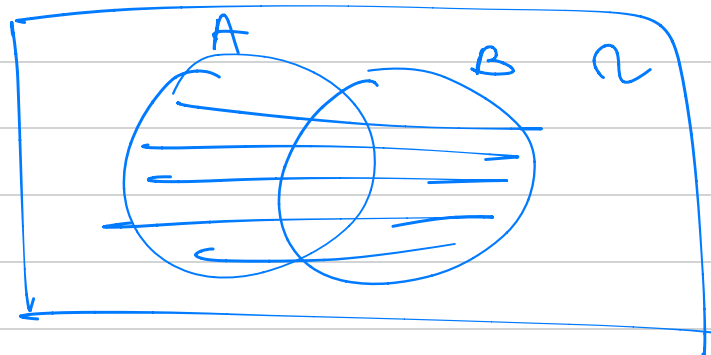
$$A_1 \cap \dots \cap A_n = \{x \in \Omega \mid x \in A_i \text{ for all } i \in [n]\}$$



$A \cap B$.



$A \cup B$.



Properties of set operations

a) $(A^c)^c = A$

b) $A \cap B = B \cap A$

c) $A \cup B = B \cup A$

} commutative

d) $(A \cup B) \cup C = A \cup (B \cup C) = A \cup B \cup C$

$(A \cap B) \cap C = A \cap (B \cap C) = A \cap B \cap C$

associative {

e) $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$

distributive law {

$A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$

Exercise:

Show that

$A \cap (B \cup C)$ is not equal to $(A \cap B) \cup C$

(come up with example A, B, C)

De Morgan's law

f) $(A \cup B)^c = A^c \cap B^c$

$(A \cap B)^c = A^c \cup B^c$

$\bigcup_{i=1}^n A_i$

$= A_1 \cup A_2 \dots \cup A_n$

Exercise { $\left(\bigcup_{i=1}^n A_i \right)^c = \bigcap_{i=1}^n A_i^c$, $\left(\bigcap_{i=1}^n A_i \right)^c = \bigcup_{i=1}^n A_i^c$

$$S_1 = (A \cup B)^c \Rightarrow A^c \cap B^c = S_2. \text{ To show } S_1 = S_2$$

$$\begin{array}{l} \textcircled{a} \text{ If } x \in S_1 \Rightarrow x \in S_2 \text{ then } S_1 \subseteq S_2 \\ \textcircled{b} \text{ if } x \in S_2 \Rightarrow x \in S_1 \text{ then } S_2 \subseteq S_1 \end{array}$$



$$S_1 = S_2.$$

$$\textcircled{a} \quad x \in (A \cup B)^c \Leftrightarrow x \notin (A \cup B)$$

$$\Leftrightarrow x \notin A \text{ and } x \notin B.$$

$$\Leftrightarrow x \in A^c \text{ and } x \in B^c$$

$$\Leftrightarrow x \in A^c \cap B^c.$$

$$\Rightarrow \begin{array}{l} (A \cup B)^c \subseteq A^c \cap B^c \\ (A \cup B)^c \supseteq A^c \cap B^c \end{array}$$

Cardinality of a set:

(Ref: Principles of
Mathematical Analysis
by W. Rudin
Chapter 2)

$$\text{If } S = \{x_1, x_2, \dots, x_n\}$$

$$|S| = n \text{ here.}$$

① S is finite when it has
finite # of elements.

$$S = \{1, 2, \dots, 6\}$$

$$\mathbb{Q} \cap [0, 1] \rightarrow \text{countable}$$

$\rightarrow$ rational numbers

$$[0, 1] \rightarrow \text{uncountable.}$$

② infinite :

Examples: $\Omega = \{H, TH, TTH, \dots\}$

$$\mathbb{Z} = \{-2, -1, 0, 1, 2, \dots\}$$

$$\mathbb{N} = \{1, 2, 3, \dots\}$$

rational $\leftarrow \mathbb{Q}$, $\mathbb{R} \rightarrow$ real.